When Software Fabricates Assurance: Seven Design Principles for Trustworthy Governance, Risk and Compliance Systems

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Name of the Track: Track 1 – Information Systems: Theory and Practice (Digital Ethics and Policy: Regulation and Compliance)

Type of Submission: Ongoing research (design science research; artifact and technical evaluation completed, naturalistic evaluation pending)

Declaration on the use of generative AI

Claude (Claude Opus 5.5, Anthropic) was used under the authors' direction to assist the audit, implement much of ERMA-GRC, draft the protocol, write and run the test scripts, propose outcome codings, produce the figures, support the literature search, and draft text. The authors set the questions, approved each stage including the protocol before execution, verified sources and results, revised the text and take full responsibility. No AI tool is an author.

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Abstract

Purpose

Boards, auditors and regulators rely on governance, risk and compliance (GRC) information systems for evidence that risks are identified, rated, approved and controlled. Research on these systems has examined their adoption and value. We found no study asking whether the evidence they present can itself be trusted. A code and runtime audit of an existing risk management application found that the software invented risk ratings, discarded approvals, accepted unauthenticated authority and reported failed database writes as successes. This study asks which design failures compromise the integrity of assurance evidence. It also asks which design principles prevent them, and how well an instantiation of those principles performs against the original system.

Design/Methodology/Approach

This is a design science study. We read the code of an existing risk management application, ran it, and recorded seven distinct ways in which it failed. Each failure was traced to theory that already explains it, from integrity models, fail-safe defaults and complete mediation to separation of duties, tamper-evident logging and the critique of audit as ritual, and a design principle was drawn from that theory. We built a new system, ERMA-GRC, on these principles. The old and new systems then faced 23 scenarios over the same stored data, under a protocol time-stamped in the project repository beforehand. Eighteen scenarios come from the audit, four go beyond it, and one was meant to fail. All deviations are reported.

Findings

Out of the 18 possible cases resulting from the audit test, the old system blocked the failure of security in 4 of these. On the other hand, ERMA-GRC blocked all 18 of the possibilities. ERMA-GRC blocked writes outside a session, blocking a person from approving his/her record, stamping the audit lines for signed-in users and not the names that are entered, detecting an edit of an entry and reporting unavailability of the database as failure. In addition to these audit test conditions, ERMA-GRC blocked 3 out of the 4 failures possible, where in two cases it was two simultaneous approvals for the same record. Storage results were collected from the database simulator, rather than the system itself. There were 2 scenarios when the new design did not block the failure, and these are mentioned below. Although one risk indicator displayed the right state on the screen, a client can connect to the server and set any state that he/she wishes. Furthermore, a person having access to the audit key can modify the audit trail offline, without detection after server restart. The principles also rely on each other since the reliability of audit depends on separation of duties.

Originality/Value

We treat the integrity of assurance evidence as something a GRC system's design can protect or destroy, rather than as an organisational outcome. This paper provides a taxonomy of failures observed in a working system, seven principles together with two testable propositions towards a design theory, and a scenario protocol that others can rerun. They can use the principles as an action list. The two issues in the design give rise to the following three exercises: the anchoring of the audit trail outside of the system; the computation of all metrics in the server, and testing of the principles using practitioners.

Keywords: assurance integrity; GRC information systems; design science research; enterprise risk management; tamper-proof audit trail; separation of duties

1. Introduction

Risk registers, risk maps, assessments signed off and audit trails are mainly in governance, risk and compliance (GRC) information systems. To boards, auditors and regulators, that is sufficient proof of risk management (Power, 1997). The sufficiency of the proof lies in its documentation.

We started the research with an audit of the code and run-time behaviour of a GRC application documented as ready for deployment. The application displayed assurance with nothing behind it. Unrated risks received scores computed from the related record's identifier or a default loss value. Any further save reverted a signed-off assessment to draft status. The claimed immutable audit trail was stored in the browser, where any user could tamper with it. On failure of writing data, users were assured of successful operation anyway. Everyday design decisions turned the system into the cause of the risks instead of the risk mitigation mechanism.

As for literature on the problem, there were no such pieces in our review. GRC IS research deals with the reasons for implementing the systems and benefits companies receive from that (Papazafeiropoulou & Spanaki, 2016; Racz et al., 2010). Enterprise risk management literature mentions that assurance may turn into ritualistic practice (Power, 2009). Suitable security-related literature offers solutions too, but only in the general case (Clark & Wilson, 1987; Saltzer & Schroeder, 1975). There is no such research linking design mistakes of applications to corrupted assurance evidence and testing design concepts against a practical system.

We therefore ask which design failures in a GRC information system compromise the integrity of its assurance evidence (RQ1); which theory-grounded design principles address them (RQ2); and to what extent an instantiation of these principles prevents the failures compared with the original system (RQ3). Our design science research approach yields a failure taxonomy, seven principles and a new artifact, ERMA-GRC, tested through time-stamped scenarios. The next section covers the move from literature review through to method, results, implications and limitations.

2. Literature Review

2.1 Conceptual Background

ERM is the systematic framework used by the enterprise to identify, assess, manage and monitor the risks. GRC information system integrates all the activities of ERM and documents it (Racz et al., 2010). Documentation of this risk management process is crucial because it assures the board, auditor and regulator that the risks are being properly managed.

Assurance integrity is an attribute of the GRC information system that makes every assurance provided by the GRC information system either recorded as an action for which someone is answerable or recorded as the lack of a record of such action. It is an attribute of the system, not of the risks.

2.2 Review of Previous Studies

Organisational research shows that ERM is enacted rather than adopted. Arena et al. (2010) trace how risk rationalities, experts and technologies combine differently across firms; Mikes (2009) identifies distinct "calculative cultures"; Mikes and Kaplan (2015) argue that a single ERM template is itself a risk; and Meidell and Kaarbøe (2017) show that a risk function's tools shape its influence. Power (1997) warns that verification rituals can yield auditability rather than assurance, and Power (2009) calls ERM at its weakest a "risk management of nothing". Technology therefore helps produce risk, so its design merits study.

GRC information systems research has defined the field (Racz et al., 2010), identified implementation success factors (Papazafeiropoulou & Spanaki, 2016), and described an adoption lifecycle and value drivers (Spanaki & Papazafeiropoulou, 2016). It asks how these systems are adopted and what value they deliver, not whether their evidence can be trusted.

The artifact's mechanisms exist in generic form: well-formed transactions, integrity verification and separation of duty (Clark & Wilson, 1987); fail-safe defaults and complete mediation (Saltzer & Schroeder, 1975); separation of duties in workflows (Botha & Eloff, 2001); and tamper-evident logs, using hash chains with evolving keys (Schneier & Kelsey, 1999) or history trees kept honest by external auditors (Crosby & Wallach, 2009). Continuous assurance monitors business-process controls through analytics (Alles et al., 2006; Vasarhelyi et al., 2004), focusing on monitored processes rather than faults in the monitoring software. Table 1 summarises the seven closest studies; five were assessed from abstracts only.

Table 1. Summary of Selected Literature

Author(s)	Context	Methodology	Key Findings	Gap Identified
Racz et al. (2010)	Integrated GRC	Literature analysis, definition	A frame of reference for GRC research	Not its focus: artifact design and evaluation
Papazafeiropoulou & Spanaki (2016)	GRC IS implementation	Literature review and expert interviews	Implementation characteristics and success factors	Integrity of the evidence the systems produce
Spanaki & Papazafeiropoulou (2016)	GRC IS adoption	Qualitative study	Adoption lifecycle; integration, optimisation and information as value drivers	Not its focus: trustworthiness of that information
Arena et al. (2010)	ERM in organisations	Comparative case study	ERM varies with its rationalities, experts and technologies	Not its focus: how a technology's design distorts risk information

Author(s)	Context	Methodology	Key Findings	Gap Identified
Alles et al. (2006)	Continuous control monitoring	Pilot implementation (Siemens)	Business-process controls can be monitored continuously	Not its focus: design faults in the monitoring software
Botha & Eloff (2001)	Workflow access control	Conceptual model	Static and dynamic separation-of-duties enforcement	Not its focus: approval semantics of risk assessments
Crosby & Wallach (2009)	Secure logging	Data structure and analysis	Efficient tamper-evident logs, provided auditors hold commitments	Business workflows and other failure types

2.3 Research Gap

Within the web-indexed literature our search reached, with every DOI verified, no study examines how a GRC application's design can itself corrupt the assurance evidence it presents: by fabricating values, overwriting decisions, accepting unauthenticated authority or reporting failures as success. GRC research stops at adoption and value, while mechanism research does not address the evidence that risk and control self-assessment produces: ratings, approvals, appetite exposure and indicator status. The gap is therefore not a missing mechanism but missing design knowledge that maps application failures to corrupted evidence and tests principles against a real system.

2.4 Research Framework

The study links observed failure modes (RQ1), theory-grounded principles (RQ2), their mechanisms, and scenario outcomes testing whether the failures are prevented (RQ3). These contribute components of a design theory (Gregor & Jones, 2007): purpose and scope, constructs, principles of form and function, and testable propositions (Section 5). Each principle pairs an aim with a mechanism and a kernel-theory rationale (Gregor et al., 2020).

3. Methodology

3.1 Research Design

Our approach is design science (Hevner et al., 2004), following the six steps of Peffers et al. (2007): (1) problem identification, the audit (RQ1); (2) objectives; (3) design of seven principles, instantiated in ERMA-GRC (RQ2); (4) demonstration on Oracle Autonomous Database; (5) evaluation (RQ3); and (6) communication. The evaluation is one artificial, summative episode of the Technical Risk and Efficacy strategy of the Framework for Evaluation in Design Science (FEDS; Venable et al., 2016).

3.2 Data Collection

Evidence came from the audit and a scenario experiment, its protocol time-stamped in the project repository (not a public registry) before execution. The original system (baseline) and ERMA-GRC ran against one database stand-in with identical rows (Figure 1), one recorded run per scenario, keeping full evidence.

3.3 Measures and Instruments

Eighteen of the 23 scenarios operationalise the failure modes, four test the principles beyond them, and one was expected to fail. Runs were coded PREVENTED, OCCURRED or not applicable against criteria registered in advance, reading stored state from the stand-in and display outcomes from screen text.

3.4 Data Analysis

We count PREVENTED scenarios per failure mode; deterministic outcomes need no inferential statistics. All deviations are reported in a replication package available on request: the test scripts were piloted after registration; open points (e.g., an approver as editor in S2.2) were fixed before each run; the rule coding G4 by its stricter path was not registered; and S5.1 was re-run after a script fault.

Use of generative AI. Claude (Claude Opus 5.5, Anthropic) was used under the authors' direction to assist the audit, implement much of ERMA-GRC, draft the protocol, write and run the test scripts, propose outcome codings, produce the figures, support the literature search, and draft text. The authors set the questions, approved each stage including the protocol before execution, verified sources and results, revised the text and take full responsibility. No AI tool is an author.

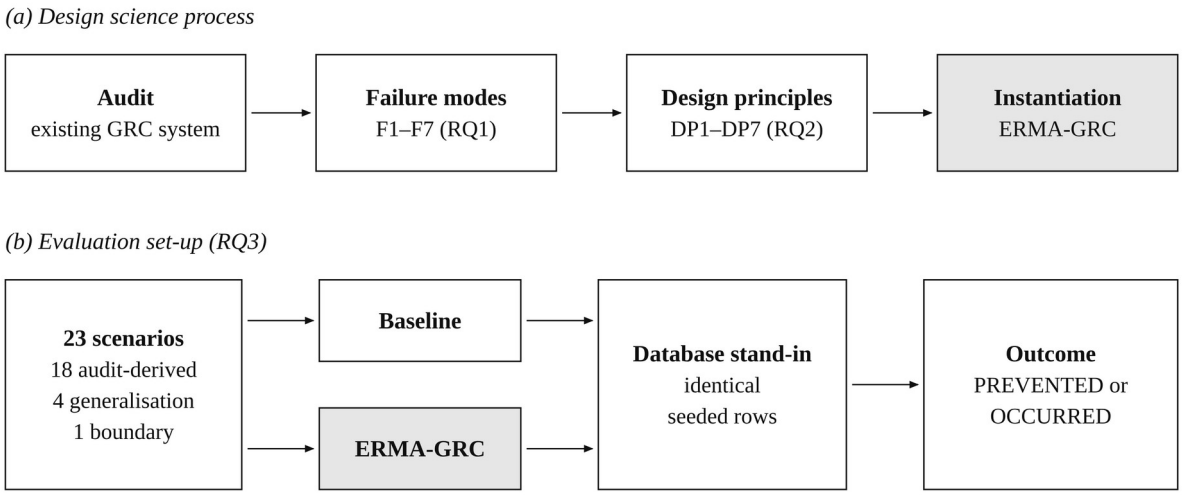


Figure 1. Design science process and evaluation set-up: the audit yields F1–F7, which inform DP1–DP7, instantiated in ERMA-GRC; both systems run the 23 scenarios against one database stand-in.

4. Results and Discussion

4.1 Results

RQ1 and RQ2. Table 2 maps each failure mode to its principle, kernel theory and mechanism in ERMA-GRC.

Table 2. Failure Modes, Design Principles, Kernel Theories and Mechanisms

Failure mode (RQ1)	Design principle (RQ2)	Kernel theory	Mechanism in ERMA-GRC
F1 Fabricated assurance values	DP1 Evidence provenance	Clark & Wilson (1987); Power (1997)	No derived ratings; "Not rated" shown

Failure mode (RQ1)	Design principle (RQ2)	Kernel theory	Mechanism in ERMA-GRC
F2 Decisions overwritten by defaults	DP2 Decision preservation	Clark & Wilson (1987); Saltzer & Schroeder (1975)	Defaults only at creation; approvals only via authorised transitions
F3 Client-side or absent enforcement	DP3 Server-enforced, identity-bound separation of duties	Saltzer & Schroeder (1975); Botha & Eloff (2001)	Server session; deny by default; maker ≠ checker; conflicts refused; revocation
F4 Mutable, forgeable evidence trail	DP4 Tamper-evident, actor-bound trail	Schneier & Kelsey (1999); Crosby & Wallach (2009)	Keyed hash chain (static key); session-bound actor; visible verification
F5 Silent failure reported as success	DP5 Fail-visible operation	Saltzer & Schroeder (1975)	Errors returned; no fallback store
F6 Fragmented system of record	DP6 Single system of record	Clark & Wilson (1987)	Records in database; accounts and audit chain on server
F7 Destructive actions on fabricated records	DP1 Evidence provenance	as for F1	Empty states, no placeholder rows
F1 (metrics)	DP7 Computed, not declared, metrics	Vasarhelyi et al. (2004); Cox (2008)	Status computed from thresholds, in the client

RQ3. ERMA-GRC prevented all 18 audit-derived failures; the baseline prevented 4 (Figure 2). The baseline stored DRAFT after an approver set APPROVED, accepted writes without a session or with a self-declared administrator header, recorded an audit entry under a claimed "CEO" identity, reported success while the database was unreachable, and invented two scored risks, each with a delete button, for a cycle that had none. ERMA-GRC prevented three of the four generalisation failures: it kept one of two simultaneous approval decisions, rejected a write from a deactivated user's still-valid session, and rejected a path-traversal identifier before any database request. The baseline's 0/4 there largely repeats its F2 and F3 defects and is not independent evidence. For balance, when an editor without approval rights changed an approved risk (S2.2 variant), ERMA-GRC returned it to review by design, which the literal criterion codes as OCCURRED; the simultaneous-decision race on a record's status (G1 variant) was prevented.

Where the baseline passed. The baseline displayed a stored statement and single loss expectancy (SLE) faithfully (S1.2), showed a missing approval status as "Draft", as ERMA-GRC does, rather than Approved (S1.3), held no invented entries in a fresh audit trail (S1.4), and attributed an audit entry to the self-asserted acting persona (S4.3). S1.3 and S1.4 contradict the audit report; for S1.4, the seeding code exists but never runs on a fresh installation.

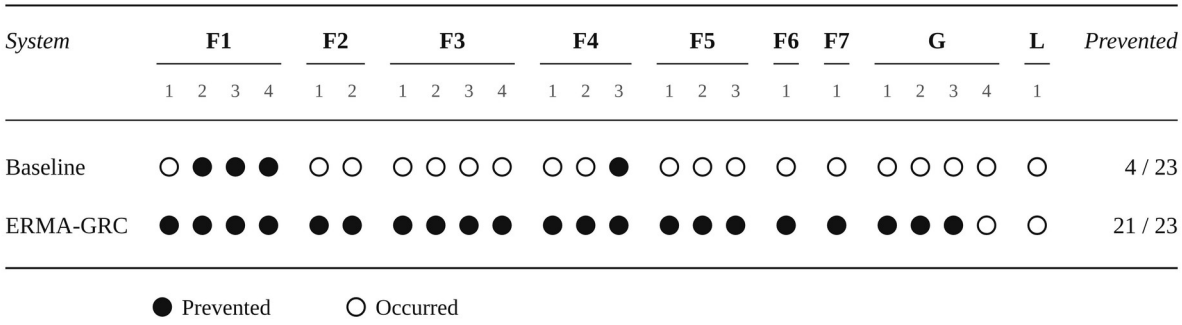


Figure 2. Outcome of every scenario for each system, grouped by failure mode (F1–F7; see Table 2), generalisation scenarios (G) and the boundary scenario (L). Within each group, columns are its scenarios in order (e.g., column 3 under F1 is S1.3). Filled circle: PREVENTED; open circle: OCCURRED.

Where ERMA-GRC failed. In G4, a key risk indicator (KRI) reading received its status by rule in the interface, but the API stored a status declared by the client, as the baseline's does. In L1, an insider holding the audit key re-chained an edited log offline, and verification passed after a restart. Table 3 contrasts two baseline failures with ERMA-GRC on the same stored data.

Table 3. What Each System Showed for Identical Records

Scenario	Database content	Baseline displayed	ERMA-GRC displayed
S1.1 Unrated risk	No likelihood, impact or SLE	Rating 1/5 × 4/5 (score 4, Moderate); SLE $40,000	Inherent and residual "Not rated"; SLE "—"
S7.1 Cycle with no risks	A cycle holding no risks	Two invented risks (scores 16 and 12), each with a Delete button	"No risks in this cycle yet"

4.2 Discussion

The failures are assurance failures. The baseline produced scores, approvals and audit entries disconnected from its stored state: a software-made "risk management of nothing" (Power, 2009). Established mechanisms counter this: fail-safe defaults and complete mediation (Saltzer & Schroeder, 1975), separation of duties (Botha & Eloff, 2001), and well-formed transactions with integrity verification (Clark & Wilson, 1987). DP1, DP2 and DP7 specialise these integrity ideas to assurance evidence: they ask whether a displayed rating, decision or metric is evidence at all.

The principles depend on each other. ERMA-GRC reads a record's creator, and hence who may not approve it, from its audit trail; in L1, rewriting one creation entry changed who was barred from approving that record. DP3 therefore rests on DP4, and DP4 on keys held outside the application. Unlike Schneier and Kelsey's (1999) evolving keys, ERMA-GRC's key is static, and without external commitments its holder can rewrite history (Crosby & Wallach, 2009).

Enforcement must sit at the server. G4 shows that a rule checked only in the client is a convention, not a control. Integrity is necessary but not sufficient: faithful records of a poorly calibrated risk matrix are still poor risk information (Cox, 2008).

5. Implications

Theoretical implications. The study recasts assurance integrity as a design property of GRC systems and contributes components of a nascent design theory (Gregor & Jones, 2007): purpose and scope, constructs, seven principles and two testable propositions. P1: separation of duties is only as strong as the integrity of the evidence trail it reads. P2: an assurance rule enforced only in the client can be defeated by any authenticated client.

Practical implications. Before purchase, buyers and auditors can test whether an unrated risk is shown as unrated, whether an ordinary save can overwrite an approval, whether authority comes from a verified session, whether the audit trail is verifiable and anchored outside the vendor's control, and whether a failed write can ever appear as a success.

Societal implications. Regulators and the public increasingly rely on compliance claims produced by software. Systems that cannot fabricate evidence make those claims more trustworthy and accountability harder to evade.

6. Conclusion, Limitations and Future Research

This study asked how a GRC system's design can corrupt its assurance evidence, and how to prevent it. An audit yielded seven failure modes, addressed by seven theory-grounded principles. ERMA-GRC, which instantiates them, prevented every audit-derived failure; the original system prevented four of eighteen. Its two failures, a metric enforced only in the client and an audit trail its key holder can rewrite, are part of the contribution.

Limitations. The evidence comes from one case. The authors built the artifact and wrote the scenarios with an AI assistant involved in every stage, from audit to outcome coding; time-stamping the protocol and reporting every outcome reduce but do not remove this bias. The evaluation was artificial: both systems ran against a stand-in derived from ERMA-GRC's own test double, so concurrency and outage results depend on its fidelity, and the display scenarios lacked a rated risk as a positive control. The literature search lacked Scopus and Web of Science, and no commercial product was compared.

Future research. Next steps are naturalistic evaluation with practitioners, multiple-case studies across GRC products, externally anchored audit commitments, server-side computation of every assurance metric, and longitudinal study of how integrity-preserving systems change the use of risk information.

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